Supplemental Data

Table S1: Eligibility criteria

Inclusion criteria	Exclusion criteria
Clinical trials or observational studies with ≥10 patients treated with anti-CD20 therapy, which assessed vaccine response to clinically available vaccines	Case reports and case series
Studies reporting vaccine response outcomes specific to patients who had received anti-CD20 therapy	Conference proceedings
Studies reporting vaccine response outcomes of seroresponse, seroprotection, seroconversion, and/or cellular response	Studies of cancer vaccines
English language reports	

Table S2.1 Database-specific search strategy for Pubmed

Anti-CD20 therapy term	Keywords searched for in abstract and title	anti-CD20 **OR** CD20 **OR** rituximab **OR** obinutuzumab **OR** ofatumumab **OR** ocrelizumab **OR** tositumomab **OR** ibritumomab **OR** Ocaratuzumab **OR** veltuzumab
		OR
	MeSH headings	("Bi20 antibody"[Supplementary Concept])
		AND
Vaccination term	Keywords searched for in abstract and title	vaccin* **OR** vaccination **OR** immunization **OR** immunisation
		OR
	MeSH headings	("vaccines"[MeSH terms]
		NOT
Exclusion term	Keywords searched for in abstract and title	"cancer vaccin*" NOT "idiotyp*"
Additional limits		Limit to English language only

Table S2.2 Database-specific search strategy for EMBASE

Anti-CD20 therapy term	Keywords searched for in abstract and title	(anti-cd20 **OR** rituximab **OR** obinutuzumab **OR** ofatumumab **OR** ocrelizumab OR tositumomab **OR** ibritumomab **OR** ocaratuzumab **OR** veltuzumab).mp
		OR
	Subject headings	(exp rituximab/ **OR** exp CD20 antibody/)
		AND
Vaccination term	Keywords searched for in abstract and title	(vaccin*.mp. **OR** immuniz*.mp.)
		OR
	Subject headings	(exp vaccination/ **OR** exp vaccine/ **OR** exp immunization/)
		NOT
Exclusion term	Keywords searched for in abstract and title	(tumor vaccine.mp. **OR** cancer vaccine.mp **OR** idiotyp*.mp)
		OR
	Subject headings	(exp tumor vaccine/ **OR** exp cancer vaccine/ **OR** exp antiidiotypic antibody/ **OR** exp idiotypic antibody/)
Additional limits		Limit to English language only Limit to clinical trials and review articles Exclude conference proceedings

Table S3: Details on risk of bias tool used for each study

Type of study	Risk of bias tool	Scoring
Cohort studies and sub-studies of trials with a comparison group	Newcastle-Ottawa assessment tool[1,2]	**Good quality**: 3 or 4 stars in selection domain AND 1 or 2 stars in comparability domain AND 2 or 3 stars in outcome/exposure domain **Fair quality**: 2 stars in selection domain AND 1 or 2 stars in comparability domain AND 2 or 3 stars in outcome/exposure domain **Poor quality**: 0 or 1 star in selection domain OR 0 stars in comparability domain OR 0 or 1 stars in outcome/exposure domain
One-arm cohort studies	Joanna Briggs Institute Critical Appraisal Checklist for Cohort Studies[3]	Low risk of bias: 6 out of 9 points
Randomized controlled trials	Cochrane Risk of Bias Scale[4]	Utilization of algorithm to define low risk

Table S4: Response definitions for vaccinations by study

Study	Response definition	SP	SC	SR
T-cell dependent vaccinations				
Pandemic influenza				
Bedognetti 2012	Seroprotection: Antibody titer ≥ 40 Seroconversion: ≥4-fold increase of antibody titer	X	X	
Berglund 2014	Seroprotection: Antibody titer ≥ 40 Seroconversion: ≥four-fold increase in post-vaccination titre, or if prevaccination titre was <10 an increase ≥40	X	X	
de Lavallade 2011	Seroprotection: titers ≥1:32 Seroconversion: pre-vaccination titer <1:8 and post-vaccination titer ≥1:32, or pre-vaccination titer ≥1:8 and an increase in the titer by a factor of four or more	X	X	
Hottinger 2012	Seroprotection: titer ≥1:40 Seroconversion rate: titer ≥1:40 and a four-fold increase in GMT	X	X	
Ide 2014	Seroresponse: ≥ 4-fold rise Seroprotection: titer ≥ 1:40 Seroconversion: baseline titer < 1:10 and postvaccination titer ≥ 1:40 or baseline titer ≥ 1:10 and ≥ 4-fold rise [not reported for rituximab-specific data]	X	X	X
Issa 2011	Seroprotection: Titre ≥1:40	X		
Kapetanovic 2014	Seroprotection: Titre ≥1:40 Seroconversion: negative prevaccination serum and postvaccination titer ≥40 or a ≥4-fold increase in HI titer	X	X	
Kim 2013	Seroprotection: Titre ≥1:40 Seroconversion: at least a four-fold increase in HI antibody levels or seroconversion after vaccination compared with titers before vaccination	X	X	
Mackay 2011	Seroprotection: Titre ≥1:40 Seroconversion: antibody titre ≥4-fold baseline	X	X	
Monkman 2011	Seroprotection: Titre ≥1:80 Seroconversion: antibody titre ≥4-fold baseline	X	X	
Muller 2013	Significant levels of specific antibody responses (titer >1:8); response increase (≥2 titration steps or minimum fourfold increase)			X
Villa 2013	Seroprotection: titer ≥ 1:40 Seroconversion: baseline titer < 1:10 and postvaccination titer ≥ 1:40 or baseline titer ≥ 1:10 and ≥ 4-fold rise	X	X	
Yri 2011	Seroprotection: titer ≥ 1:40	X		
Seasonal influenza				
Arad 2011	Either fourfold or more rise in the titer of HI antibodies, or a rise from a non-protective baseline level (<1/40) to ≥1/40.	X	X	
Bar-Or 2020	Seroprotection: HI titers ≥40 Seroconversion: ≥4-fold increase in HI titers after vaccination (in patients with prevaccination HI titers ≥10) or postvaccination HI titers ≥40 (in patients with prevaccination HI titers <10)	X	X	
Bedognetti 2011	Seroprotection: HI titer ≥40 Seroconversion: ≥4-fold increase in HI titer from a nonnegative pre-vaccination titer (i.e., ≥40) or increase from <10 to ≥40 in previously seronegative subjects.	X	X	
Bedognetti 2012	Seroprotection: antibody titer ≥ 40 Seroconversion: ≥4-fold increase of antibody titer after vaccination *conjugated seasonal influenza	X	X	
Berglund 2014	Seroprotection: HI titers ≥40 Seroconversion: ≥4-fold increase in HI titers after vaccination (in patients with prevaccination HI titers ≥10) or postvaccination HI titers ≥40 (in patients with prevaccination HI titers <10)	X	X	
Cho 2017	HI titre ≥40	X		
Eisenberg 2013	Two-fold increase to at least one serotype at the two-month time point after vaccination			X
Oren 2008	Four-fold or more rise in titre, or a rise from a non-protective baseline level of <1/40 to ≥1/40 in HI antibodies 4 weeks after vaccination		X	
Richi 2019	Cut-off value: 10 AU. Samples considered positive if absorbance value at least 15% over the cut-off: ≥11.5 AU and negative if absorbance value was lower than 15% below the cut-off: <8.5 AU. Samples with an absorbance value of up to 15% above or below the cut-off were considered neither positive nor negative. Nonetheless, to avoid bias that could lead to better results we considered negative any value under 11.5 AU. Final seropositivity rates against Ag A and Ag B after vaccination were calculated.	X		
van Assen 2010	≥4-fold titer rise resulting in a postvaccination level of ≥40 (seroconversion), and titer rise to ≥40 (seroprotection)	X	X	

Tetanus				
Albert 2008	2 fold rise in titre			X
Bar-Or 2020	Anti-TT IgG antibody titer ≥0.2 IU/mL in patients with prevaccination titers <0.1 IU/mL or ≥4-fold increase in antibody titer in patients with prevaccination antibody titers ≥0.1 IU	X	X	
Bingham 2010	Anti-TT IgG antibody titer ≥0.2 IU/mL in patients with prevaccination titers <0.1 IU/mL or ≥4-fold increase in antibody titer in patients with prevaccination antibody titers ≥0.1 IU		X	
Buhler 2019	Seroprotection: IgG >0.1 IU/ml, Response: Multifold increases in concentrations (≥4 reported)	X	X	
Colucci 2019	Seroprotection: 0.6IU/mL	X		
Horwitz 2004	Absolute values or an observed 4-fold or higher increase over baseline			X
Mustafa 2020	2-fold increase, and must be in protective range (0.1 IU/mL)	X	X	
Pescovitz 2011	2- fold increase in titer			X
Puissant-Lubrano 2010	Seroprotection: 0.15 IU/mL for tetanus toxoid. Response: 4-fold increase of ratio	X		X
Shah 2015	Seroconversion in a seronegative individual, a 3-fold geometric mean fold rise of the IgG geometric mean concentration		X	
van der Kolk 2002	NR (fold increases reported; seroconversion defined using other reported definitions of anti-TT IgG antibody titer ≥0.2 IU/mL in patients with prevaccination titers <0.1 IU/mL or ≥2-4-fold increase in antibody titer in patients with prevaccination antibody titers ≥0.1 IU)		X	
Diptheria				
Buhler 2019	Seroprotection: IgG >0.1 IU/ml	X		
Mustafa 2020	2-fold increase, and must be into protective range (0.1 IU/mL)		X	
Pescovitz 2011	2-fold increase in titer			X
Shah 2015	Seroconversion in a seronegative individual, a 3-fold geometric mean fold rise of the IgG geometric mean concentration		X	
Pertussis				
Shah 2015	Seroconversion in a seronegative individual, a 2-fold geometric mean fold rise of the IgG geometric mean concentration		X	
Small 2009	Seroprotection: >5 IU/mL; seroresponse: ≥2-fold increase	X		X
Hemophilis influenza B				
Horwitz 2004	Antibody titers were graded as protective based on absolute values or an observed 4-fold or higher increase over baseline.			X
Nazi 2013	4-fold increase in antibody concentration from baseline within the first month after vaccination. A specific IgG titer of 1 μg/mL or higher was considered protective		X	
Shah 2015	Seroconversion in a seronegative individual, a 3-fold geometric mean fold rise of IgG GMT		X	
Hepatitis B				
Avivi 2019	Titres >10 IUμ/L	X		
Jaffe 2006	Patients who tested seronegative prior to vaccination were considered responders if their postvaccine serum testing showed absorbance above or equal to the kit cutoff value	X	X	
Richi 2020	Seroconversion when titer >10 mIU/mL Non-responder when, despite two correctly administered full courses of HBV vaccine, failed to cross an anti-HBs threshold of 10 mIU/mL, determined at least 4 weeks after the last vaccine dose	X	X	
Hepatitis A Virus				
Pescovitz 2011	Seroprotection: titres >20 mIU/mL	X		
van der Kolk 2002	Seroprotection titres >20 mIU/mL	X		
Polio virus				
Jaffe 2006	3-fold or greater rise in titer against all 3 polio serotypes after receipt of 3 IPV vaccinations given at 2-month intervals			X
Shah 2015	Response: seroconversion in a seronegative individual, a 3-fold geometric mean fold rise of the IgG geometric mean concentration for all 3 serotypes			X
van der Kolk 2002	NR; defined using above definitions			X
Pneumovax conjugate (PCV-7)				
Crnkic Kapetanovic 2013	Response: ≥2-fold increase in pre- and post-vaccination antibody levels			X
T-cell independent vaccinations				
Pneumococcal polysaccharide vaccine (PCV-23)				
Albert 2008	2 fold rise in titre			X
Bar-Or 2020	2-fold or >1 μg/mL increase in serotype-specific antibody titers			X
Berglund 2014	2-fold or greater increase in antibody levels together with IgG levels ≥1.0 μg/ml. A response to at least seven (50%) of the 14 pneumococcal serotypes was regarded as seroconversion response.	X	X	
Bingham 2010	2-fold increase or an increase of >1 μg/ml from prevaccination levels			X
Horwitz 2004	Antibody titers were graded as protective based on absolute values or an observed 4-fold or higher increase over baseline			X
Mustafa 2020	If < 1.3 mg/mL, 2-fold increase to > 1.3 mg/mL OR 4-fold increase	X	X	

	If > 1.3 mg/mL, 2-fold increase Responses must be demonstrated by 70% of serotypes			
Nazi 2013	4-fold increase in antibody concentration from baseline within first month after vaccination			X
AU: absorbance units; GMT: geometric mean titres; HI: hemagglutination inhibition; NR: not reported; SC: seroconversion; SP: seroprotection; SR: seroresponse; TT: tetanus toxoid				

Table S5: Results of risk of bias assessments for individual studies

Study	Overall risk of bias
Albert 2008[5]	High
Arad 2011[6]	High
Avivi 2019[7]	High
Bar-Or 2020[8]	Low
Bedognetti 2011[9]	Low
Bedognetti 2012[10]	High
Berglund 2014[11]	Low
Bingham 2010[12]	Some concerns
Buhler 2019[13]	Low
Cho 2017[14]	Low
Colucci 2019[15]	High
Crnkic Kapetanovic 2013[16]	Low
de Lavallade 2011[17]	Low
Eisenberg 2013[18]	Low
Horwitz 2004[19]	High
Hottinger 2012[20]	Low
Ide 2014[21]	Low
Issa 2011[22]	Low
Jaffe 2006[23]	Low
Kapetanovic 2014[24]	Low
Kim 2013[25]	High
Mackay 2011[26]	Low
Monkman 2011[27]	Low
Muller 2013[28]	Low
Mustafa 2020[29]	Low
Nazi 2013[30]	Low
Oren 2008[31]	Low
Parrino 2017[32]	Low
Pescovitz 2011[33]	Low
Puissant-Lubrano 2010[34]	Low
Richi 2019[35]	Low
Richi 2020[36]	Low
Shah 2015[37]	Low
Small 2009[38]	Low
van Assen 2010[39]	Low
van der Kolk 2002[40]	High
Villa 2013[41]	Low
Yri 2011[42]	High
See Table S3 data for details of scoring methods used	

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